October 19

Last time, we started talking about ways to assess model quality. One important aspect to consider is whether the fitted model is successful at predicting y values for a given x value.

is fitted model successful at predicting y from x?

To answer that question, we could consider the standard error of the residuals. This quantifies how far away the actual observed value y_i tends to be from the fitted value $\hat{y}_i$, on average.

$\sqrt{\hat{\sigma}^2}$: how far away y_i tends to be from $\hat{y}_i$

Another popular measure for assessing linear models is R^2.

another measure : R^2

For simple linear regression (one predictor variable), this equals the square of the correlation between x & y, in the observed sample data.

for simple linear regression : $R^2 = (r_{xy})^2$

↖ sample correlation of x & y

The minimum possible value of R^2 is 0, which signals no linear relationship between the x and y variables.

Larger values of R^2, up to a maximum of 1, signal

stronger linear relationships between x and y.

$0 \leq R^2 \leq 1$ larger value → stronger linear relationship

R^2 also has another interpretation: it is the percentage of the total variance in the observed y values that is explained by the fitted model.

$$R = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} = 1 - \%\ \underline{\text{unexplained}}\ \text{variation in } y = \%\ \text{explained variation in } y$$

If a model explains more of the variation in y values, it is more useful for predicting y values.

higher % explained variation → model more useful for predicting y

A low R^2 value tells us that the model does not explain much of the variability in y values. This can occur for multiple reasons. One possibility is that x and y have a linear association, but it is weak.

reasons for low R^2
- weak linear relationship
- wrong model

Another possibility is that the relationship between x and y is non-linear. If the relationship is non-linear, a linear model could do a very poor job of predicting y. R^2 and the correlation, r_{xy}, measure the strength of linear relationships

This leads into the broader topic of potential problems with linear regression models. When we chose to fit a linear regression model, we made several key assumptions about the true population relationship between x and y.

what if linear regression assumptions aren't true?

- consequences?
- how to detect?
- how to remedy?

For each of those assumptions, we will consider the consequences of the assumption being violated, how to detect a violation, and what we can do to remedy it.

The first, most important assumption is that the true population relationship between x and y is linear. If that isn't true, then fitting a linear regression model is generally not a good choice. The consequence, essentially, is that all of our conclusions would be wrong.

① linearity $y = \beta_0 + \beta_1 x + \varepsilon$

Consequences of violation

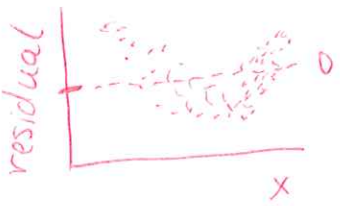
- $\hat{\beta}_0, \hat{\beta}_1$ estimates not valid
- inference not valid
- predictions can be poor

The least squares estimates of the slope and intercept are no longer valid (there ARE no true β_0, β_1 to estimate).

Any confidence intervals or hypothesis tests related to the model would not be valid.

Depending on how badly the linearity assumption is violated (ie. how different the true relationship is from a line), predictions made by the model can be arbitrarily poor.

detect via:
* x, y scatter plot
* $x, \hat{e}_i$ scatter plot



good: residuals scattered evenly above & below 0 for all x values

bad: systematic pattern in residual values

Nonlinearity can be detected by looking at a scatterplot of the data, x and y, or a scatterplot of the residuals $(y_i - \hat{y}_i)$ vs. the x values.

In general, the plotted residuals should show no obvious pattern, if the modeling assumptions hold. The random error terms in the population model are supposed to be random values from the same distribution. Regardless of the x value, they should be spread out evenly above and below zero. If the fitted model consistently underestimates (or overestimates) y for certain values of x, then the model does not match the true form of the relationship between x and y. What can we do differently?

remedy: non-linear transformation of x, y, or both

One thing we can do, with the set of methods we know so far, is to transform one or both of the variables. Then, fitting a linear model to the transformed variables can capture a non linear relationship between the original x and y.

fit linear model to transformed data

. There are many different transformations you could try - the decision depends on the association between x and y in the original scatterplot. Some of the most common choices are to replace x with log(x), or y with log(y), or squaring or taking the square root of a variable.



popular transformations:
- log(y) and/or log(x)
- $\sqrt{x}$, x^2, $\sqrt{y}$, y^2

② $Var(\varepsilon_i) = \sigma^2$ for all error terms (constant variance)

Another assumption we made is that the error terms in the true population model are all coming from a distribution with the same variance. If the error terms tended to be more spread out for larger values of x, that would violate the assumption of constant variance.

consequences of violation
- inference not valid
- $\hat{\beta}_0, \hat{\beta}_1$ estimates still valid
- ok to use model for prediction

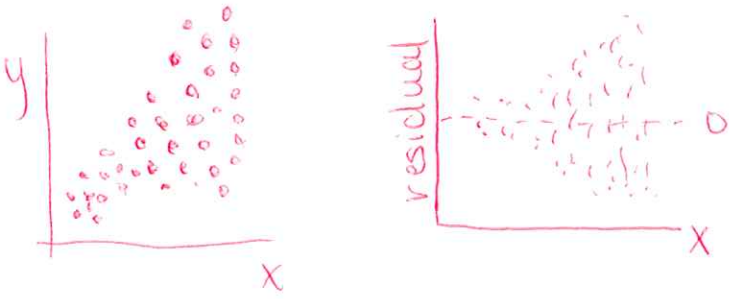
If the assumption is violated, any confidence intervals or hypothesis tests related to the model will be incorrect, because they rely on the value of σ^2.

The estimates of the slope and intercept can still be used as we did before, and the model can be used to make predictions.

Non-constant error variance can be detected using the same plots we used for linearity: scatterplot of the data,

and scatterplot of residuals vs. x.

detect via:
x_i, y_i scatter plot
$x_i, \hat{e}_i$ scatter plot



The warning sign to look for is y values, or residuals, being more spread out vertically for certain values of x, in a noticeable pattern.

good: residuals same size for all x

bad: residuals larger (magnitude) for certain x

One common problem is residuals becoming larger in magnitude for larger values of x, but this is not the only example. Residuals being more spread out for small values of x, or [redacted] for medium values, would also indicate a violation of the assumption.

remedies
- transform y (maybe)

What can we do differently? Transforming the response variable y sometimes helps. But if the relationship between x and y looks linear, transforming variables can cause non-linearity problems.

OR: If we only want to estimate the model coefficients and make predictions, but not perform inference, we don't have to do anything about the non-constant error variance.

(3) . error terms $\varepsilon \sim$ Normal

Another assumption we made about the error terms in the true population relationship is that they are normally distributed.

Confidence intervals and hypothesis tests rely on this assumption, so when the assumption is not met, they give incorrect results. However, in large samples, this usually has only a small effect, and the results are close to correct. The exception is that prediction intervals for an individual observation of y at a given x value will be incorrect even for large samples.

consequences of violation
- inference not valid
- in large samples, inference for β_0, β_1, and $\beta_0 + \beta_1 x$ close to correct
- prediction interval for y given $x = x^*$ always incorrect
- $\hat{\beta}_0, \hat{\beta}_1$ estimates still valid.

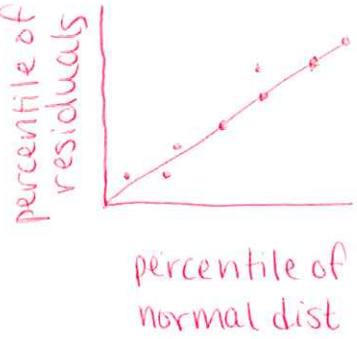
The least squares estimators of the slope and intercept are still valid, and the model can be used to make predictions (just not prediction intervals).

One way to check for non-normality is to make a histogram of the residuals, and compare to a normal distribution.

There is also a plot called a "Q-Q plot" (Q for "quantile"). This type of plot compares the percentiles of the set of residuals to the percentiles of a normal distribution.

detect via:

* histogram of residuals
* Q-Q plot of residuals



good: points on diagonal line

If the normality assumption is satisfied, the Q-Q plot will consist of points on a diagonal line. Points deviating from the line indicate potential non-normality. (You do not need to know how to make this plot from scratch.)

What can we do if the normality assumption is violated?

remedies

* sometimes: transform y
* large samples: ignore it (but don't use prediction intervals)

Transforming the y variable alleviates this, in some instances. Otherwise, in large samples it is usually alright to use the model as-is for everything except prediction intervals. In small samples, we should be skeptical of confidence intervals and hypothesis tests.

in small samples, be skeptical of all inference

The last of our core linear regression assumptions is that the error terms in the true population relationship are independent of each other (so, no correlation).

④ error terms ε independent

relevant for ordered observations

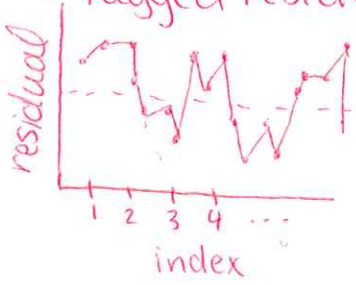
This issue is relevant when the data points in our sample have sequential structure - for instance, measurements collected over time, in order. If your data points are not ordered in a meaningful way, this is not an assumption we typically try to assess.

- consequences of violation
  - inference not valid
  - $\hat{\beta}_0, \hat{\beta}_1$ estimates still valid

If the error terms are correlated, then inference based on the model is not valid. Confidence intervals and hypothesis tests will not give correct results. The least squares estimators of the slope and intercept are still valid, and the model can be used to make predictions.

detect via:

* plot of residuals vs observation index
* lagged residuals



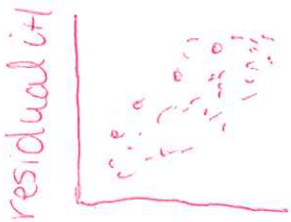
To detect this, we look at a plot of the residuals vs the observation index. That just means we plot the residuals in order: the residual for the 1st data point, then the 2nd data point, etc. If there is positive correlation, there will tend to be clusters of positive residuals followed by clusters of negative ones, alternating. Negative correlation would look like sequential data points mostly having residuals with opposite signs, ie one positive, then one negative, back & forth over and over.

bad: clusters of all-positive and all-negative

OR alternating pos-neg every time

This isn't the easiest thing to gauge visually, so it can also be helpful to plot the residuals and "lagged" residuals. This is a scatter plot where each point on the plot is of the form $(\hat{e}_i, \hat{e}_{i+1})$.

If the residuals are not sequentially correlated, this plot will look like a round blob. A diagonal shape indicates possible violations.



each point $(\hat{e}_i, \hat{e}_{i+1})$

good: round shape

bad: diagonal shape

remedy:
fit different kind of model
(haven't learned yet)

If we just want to estimate the linear relationship and make predictions, we could choose to ignore this violation.

But to go beyond that and perform inference, we would need to fit a different type of model that allowed correlated errors. This isn't something we've covered yet.

~ examples in notebook ~

more about transformations

For several different assumption violations, one of the possible approaches involves transforming variables. This is worth discussing in more detail, because transformations change the interpretation of the model coefficients.

In general, the interpretations of β_0 and β_1 apply to the transformed variables being used, but the interpretation on the original scale of the x and y values is not the same.

$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1(\sqrt{x})$

If we fit a linear regression using $\sqrt{x}$ and y, then we think y values tend to differ by $\hat{\beta}_1$ for $\sqrt{x}$ values that differ by 1 unit.

y tends to differ by $\hat{\beta}_1$ when $\sqrt{x}$ differs by 1

But this doesn't have a convenient interpretation in terms of differences in x itself.

Log transformations are somewhat special: they do have an interpretation in terms of the original x and y variables.

If we fit a linear regression model using $\log(y)$ as the response and x as the predictor...

$$\log \hat{y}_1 = \hat{\beta}_0 + \hat{\beta}_1 x_1 \qquad \hat{y}_1 = e^{\hat{\beta}_0 + \hat{\beta}_1 x_1} = e^{\hat{\beta}_0} e^{\hat{\beta}_1 x_1}$$

Then for a one unit additive difference in x, we predict...

$$x_2 = x_1 + 1 \qquad \hat{y}_2 = e^{\hat{\beta}_0} e^{\hat{\beta}_1 (x_1 + 1)} = e^{\hat{\beta}_0} e^{\hat{\beta}_1 x} e^{\hat{\beta}_1 \cdot 1}$$

The ratio of the predicted $\hat{y}$ values for x values that differ by 1 is:

$$\frac{\hat{y}_2}{\hat{y}_1} = \frac{e^{\hat{\beta}_0} e^{\hat{\beta}_1 x_1} e^{\hat{\beta}_1}}{e^{\hat{\beta}_0} e^{\hat{\beta}_1 x_1}} = e^{\hat{\beta}_1}$$

1 - unit diff in x ↓ predicted y differs by mult. factor e^{β_1}

So for a one-unit difference in x values, we expect the y values to differ by a multiplicative factor of $e^{\hat{\beta}_1}$ on average.

If we fit a linear regression model using $\log(y)$ as the response and $\log(x)$ as the predictor, and consider two predictor values x_1 and x_2 ...

$$\log(\hat{y}_1) = \hat{\beta}_0 + \hat{\beta}_1 \log(x_1) \qquad \log(\hat{y}_2) = \hat{\beta}_0 + \hat{\beta}_1 \log(x_2)$$

$$\log(\hat{y}_2) - \log(\hat{y}_1) = [\hat{\beta}_0 + \hat{\beta}_1 \log(x_2)] - [\hat{\beta}_0 + \hat{\beta}_1 \log(x_1)]$$

$$= \hat{\beta}_1 \log(x_2) - \hat{\beta}_1 \log(x_1)$$

$$= \beta_1 [\log(x_2) - \log(x_1)]$$

We can rewrite both sides of this expression using the rules of logarithms...

$$\log\left(\frac{\hat{y}_2}{\hat{y}_1}\right) = \hat{\beta}_1 \log\left(\frac{x_2}{x_1}\right)$$

Then, exponentiating both sides gives us:

$$\frac{\hat{y}_2}{\hat{y}_1} = e^{\hat{\beta}_1} * \left(\frac{x_2}{x_1}\right)$$

In this model, $e^{\hat{\beta}_1}$ relates a % difference in x to a % difference in y.

If x_2 is twice as large as x_1, then $\hat{y}_2$ is $2e^{\hat{\beta}_1}$ times as large as $\hat{y}_1$.

x differs by ↓ mult. factor of 2

predicted $\hat{y}$ differs by factor $2 * e^{\hat{\beta}_1}$

For x values that differ by a factor of 5, we expect the corresponding y values to differ by a factor of $5e^{\hat{\beta}_1}$, on average.